ICSMMEP-26 — Konuşma Metni

Eggshell–Agar Biocomposites: Development and characterisation for sustainable interior applications Betül Ünal · Cappadocia University · Mecca, 3–4 November 2026

Süre

Metnin kendisi	**11:00** · 140 kelime/dakika
27 slayt geçişi, işaret etme, nefes	~1:20
Sahnede beklenen	**≈ 12:20 – 12:45**

Provada 11:00–11:30 çıkıyorsa doğru hızdasın.

İngilizce paragraflar konuşulacak metnin kendisidir. Köşeli parantez içindeki Türkçe notlar sahne yönergesidir, okunmaz. Sayıları bilerek yazıyla yazdım.

Kalın yazılan cümleler slaytta birebir aynı duruyor. İstersen ekrandan okuyabilirsin, istersen izleyiciye söylersin — ikisi de doğru görünür. Düz yazılanlar yalnız bu metinde; onları ekrana bakmadan söyle.

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1 · Cover — 0:15

Good morning. Betül Ünal, Department of Interior Architecture, Cappadocia University. The panel in the photograph is the subject of this paper: waste eggshell bound in agar. I will give you the protocol, the characterisation, and an argument about how this class of material should be evaluated.

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2 · Background — 0:30

Eggshell is a continuous food-processing by-product, roughly a tenth of the mass of every egg, and it is predominantly landfilled. Chemically it is almost pure calcium carbonate, with a specific gravity below most mineral fillers used in construction.

[Sağdaki şemaya işaret et]

So it has two routes: landfill, or a secondary mineral resource. **The case for eggshell as a mineral phase is settled. This paper is about the binder.**

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3 · Prior applications — 0:30

This is the literature the study departs from. Eggshell has been used as a cement replacement in concrete, a stabiliser in laterite brick and in clay, a filler in asphalt, in eco-ceramic tile, in PVC, in polyurethane foam, in ceramic–metal composites, in bone scaffolds, and in wallpaper.

[Elini orta sütuna götür, aşağı indir]

But read the middle column downwards. In every case the mineral is recovered from waste while the binding phase stays cementitious, ceramic or petrochemical. **The burden is carried by the binder, and the binder is untouched.**

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4 · Agar as binder — 0:25

The binder here is agar, a thermoreversible polysaccharide from red seaweed. It sets on cooling without chemical crosslinking: no reactive hardener, no cure temperature, and at sixty-six per cent mineral loading, no coupling agent.

Its behaviour with glycerol is well mapped — but only as unfilled thin film, in tension, for packaging. **The compositional logic transfers. The loading case does not, and for an interior surface that is bending.**

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5 · Research gap — 0:30

Three gaps follow. First, the binding phase is never biological. Second, agar is characterised only as an unfilled film in tension, not as the binder of a mineral-filled cast solid in bending. Third, property sets are fragmentary, in separate studies on separate formulations.

Let me state the contribution precisely. **It is not a novel constituent. It is a documented protocol and a single coherent property set.**

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6 · Aim — 0:15

The aim, narrow on purpose: document a reproducible protocol for a fully bio-based eggshell–agar composite, and characterise it against published standards. Four objectives follow. **The fourth — deriving an evaluation criterion for this material class — is where I will spend most time.**

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7 · Scope — 0:15

Scope, set out so that nothing I say is read beyond it.

[Alttaki şeride işaret et]

The filled boxes are what was determined. The outlined boxes are what was not: durability, fire classification, life-cycle assessment, structural use.

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8 · Method — 0:15

Four stages, and the order is the method. Produce, recording every parameter. Measure against published standards. Position against two material families. Propose an evaluation criterion derived from the observed behaviour. **Properties first, application envelope second.**

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9 · Research questions — 0:15

Five questions. Two on formulation, two on benchmarking, and the fifth asks whether compressive strength is an adequate criterion for this class at all. **One to four are answered by measurement. Five is answered by argument.**

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10 · Hypothesis — 0:20

Two clauses. First: the composite can be produced repeatably under ambient conditions. Second: its performance as an interior surface is governed by deformation capacity rather than strength.

On provenance: the dissertation states a broader hypothesis framed on spatial resilience. This is its materials-engineering component, restated within the scope of these tests.

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11 · Stage 1 · Mineral phase — 0:25

Stage one. Whole shell from a single bakery, cleaned by hand, heat-treated at one hundred and twenty degrees for twenty minutes, milled, and sieved through a one hundred and fifty micrometre mesh, passing fraction only. **That particle size was chosen to match the eggshell polymer-composite literature, so the mineral phase here is comparable with that work.**

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12 · Stage 2 · Binder preparation — 0:30

Stage two, drawn as a temperature–time profile. Water and glycerol are pre-mixed before any solids. **Agar is added incrementally under continuous stirring; bulk addition forms agglomerates that do not redisperse.** Up to ninety-five degrees and held until the agar is fully in solution. Then down to eighty while viscosity develops, and the shell goes in near the end; earlier, it sediments. Fifty-five minutes in total. The set points and total time are recorded; the ramp rates are not.

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13 · Stage 3 · Casting and cure — 0:25

Stage three. Cast into silicone moulds, demoulded on day four, then seven days at twenty-three degrees. Day eleven, finished, with three shellac coats. **No thermal input after the 55-minute preparation: the entire cure is at room temperature.**

[Panele işaret et]

The limitation: the composite shrinks as water evaporates, and post-drying dimensions were not logged, so every dimension I quote is a mould dimension.

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14 · Formulations — 0:30

Two things to read here. In the compressive series, shell-to-binder is exactly two point zero zero in all three mixes. That was not planned as a control, but it means **solids concentration is the only variable — the water band shortens from C1 to C3.**

In the flexural series I declare a confound. Glycerol was halved but batch mass was held constant, so water rose to compensate.

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15 · Test programme — 0:20

Four specimen geometries, with the standard and the number of specimens under each.

I want the reporting convention on record before any numbers appear. **Mechanical values are means of three specimens; individual values were not retained, so I cannot give standard deviations.**

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16 · Compression — 0:30

The compression result, which answers question one. Solids concentration is the only variable, and compressive stress rises linearly with it, from two point five five to four point one zero megapascals.

Three points will fit almost any line, so I call this a hypothesis rather than a law. But the physical reading is clear. **Strength is governed by packing density, not by mineral content. The route to a stronger composite is to remove water, not to add shell.**

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17 · Compressive failure — 0:30

This is the failure mode. Linear to about one megapascal, then progressive loss of stiffness as the agar network yields and the mineral redistributes. Peak near ten per cent strain, no abrupt load drop.

[Fotoğrafı göster]

The cube has barrelled. It is shorter and wider, and still a single body. Failure proceeds by progressive crushing rather than by fracture. One honesty point: the strain axis assumes ten per cent maximum deformation from observation, not instrumented measurement.

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18 · Flexure — 0:30

Flexure, to ASTM D790. The decisive variable is plasticiser content. At eighteen point eight per cent glycerol the composite deformed without developing measurable resistance. Halving it gives two point three two five megapascals at a modulus of one hundred and ninety-two. **The transition is sharp, and it lies between those two values.**

Two limitations. The glycerol effect is confounded with solids concentration. And the specimen is kerfed on its tension face while the standard assumes a solid rectangle, so this modulus is a lower bound.

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19 · Flexural deformation — 0:20

I would rather show you this than ask you to take it on the numbers. The beam is deflected well past where a comparable ceramic would part. **Both test programmes gave the same result: substantial deformation with retained integrity.**

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20 · Fire, water, density — 0:30

Ignition at around ten seconds under direct flame, self-extinguishing four seconds after removal, with no lateral spread and no dripping. **This is an ignitability test; there is no Euroclass classification.**

Water: zero point seven per cent mass gain over twenty-four hours of immersion, so coating integrity is the limiting factor, not the composite. Density, one point three eight: above the fungal materials, below fired clay.

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21 · Conventional materials — 0:25

First benchmark, against what this would sit alongside in an interior. Let me be direct. One sixth the compressive strength of MDF, one twentieth that of travertine — a stone the material resembles. **On strength alone this composite is not competitive with engineered wood or stone**, and I will not claim otherwise.

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22 · Bio-based composites — 0:30

Second benchmark, and this is the one that matters. In compression, an order of magnitude above mycelium and hempcrete.

[Logaritmik eksene işaret et]

In stiffness, one hundred and ninety-two megapascals: inside the LDPE band. **Water-cast, dried at room temperature, no oven, no autoclave, no extruder — and it reaches the stiffness of melt-processed bio-polymers.** Wood is still two orders of magnitude above.

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23 · Modulus of resilience — 0:45

This brings me to question five, and the diagram is the argument.

The shaded area is the energy absorbed per unit volume before permanent damage. That area is one half sigma epsilon, and epsilon is sigma over E, so it reduces to sigma squared over two E. **Notice where the modulus sits: in the denominator.** The low modulus that counted against the material a moment ago is what raises this quantity. Here, about fourteen kilojoules per cubic metre.

Strength records the load at which a material stops performing, not whether it then fails gracefully or catastrophically — and for a demountable component, that is what governs service life.

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24 · Application envelope — 0:25

The envelope follows from the measurements. Non-loadbearing, because four point one megapascals supports nothing more. Dry interiors, because water performance depends on the coating. Away from ignition sources: self-extinguishing, but unclassified. Modular rather than sheet, because damage is then replaced one unit at a time. **No life-cycle assessment was carried out, so these are qualitative potentials.**

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25 · Limitations and further work — 0:30

The limitations, each next to the experiment that resolves it, because that pairing is the point. No dispersion data — repeat the series and retain individual values. Strain assumed — instrument it. Shrinkage unquantified — record the post-drying geometry. The kerfed section calculated as solid — test an un-kerfed one.

Every value I reported today is a lower bound. I would rather set these out than have them found.

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26 · Conclusions — 0:25

Five conclusions. A protocol that can be repeated. Packing density, not mineral content, governs compressive strength. Plasticiser content sets a brittle or a ductile response. A water-cast, ambient-dried material reaches bio-polymer stiffness without thermal processing. And strength alone is an insufficient criterion for this class: deformation capacity governs service behaviour, so strength and modulus should be reported together.

Thank you. I am happy to take questions.

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27 · References — 0:00

[Slaytı açık bırak, bir şey söyleme. Soru gelirse kaynağı buradan göster.]

Soru–cevap

Kısa tut. Sınırlılığı kabul et, sonraki deneyi söyle, sus.

"Standart sapma yok, nasıl güveneceğiz?" You are right, and I said so before the numbers. Three specimens per formulation, individual values not retained. Repeating the series with dispersion reported is the first row of my limitations table.

"Basma testi hangi standarda göre?" Cube compression, not to a named standard, and I flag that as a deviation. Flexure follows ASTM D790-17, ignitability ISO 11925-2, absorption ASTM D570.

"Ölçüler kalıp ölçüsü, gerçek gerilme değil." Correct. The composite shrinks as water evaporates and post-drying dimensions were not logged, so every value I reported is a lower bound. Recording the dried geometry is in the limitations table.

"Yivli numuneden modül hesaplanmaz." Agreed, which is why I called it a lower bound rather than a material constant. Testing an un-kerfed section is in the table.

"R^2 = 0.99 ama üç nokta var." That is exactly why I called it a hypothesis rather than a law. What I claim is the physical reading: packing density, not mineral content, governs strength.

"Dirençlilik modülü için karşılaştırma yok." No, and I cannot assemble one, because the literature reports strength and modulus in separate studies on separate formulations. I am proposing a reporting convention, not a ranking.

"Yangın mekanizması nedir?" The probable mechanism is endothermic decomposition of the carbonate fraction, diluting the flame front. I say probable because it was not verified calorimetrically.

"Nem, küf, yaşlanma?" Untested, and named as such. Durability of any kind is outside the scope of this paper.

"Ölçeklenebilir mi?" I have no data on that. What I can say is that the protocol needs a hotplate, a mould and ambient drying. No oven, no autoclave, no extruder.

"Neden agar? Nişasta ya da jelatin olmaz mıydı?" Thermoreversible setting without a hardener, and it tolerated sixty-six per cent mineral loading without a coupling agent. Comparing binders is a separate study, and a fair one to ask for.

"Yangın sınıflandırması olmadan iç mekanda nasıl kullanılacak?" It cannot be specified without one, and I say so on the application slide. What I report is ignitability behaviour, which is the precondition for seeking a classification, not a substitute for it.

Provaya dair

- Yüksek sesle, kronometreyle oku. Metnin kendisi 11:00; sahnede duraklarla

12:20–12:45.

- dakikayı geçersen **hızlanarak değil, cümle atarak kısalt.** Sırasıyla:
- Slayt 21, *"a stone the material resembles"* ara cümlesi.
- Slayt 24, son cümle (LCA kaydı).
- Slayt 17, *"One honesty point…"* cümlesi — soru gelirse söylersin.
- Slayt 12, *"bulk addition forms agglomerates…"* ara cümlesi.
- dakikanın altına düşersen slayt 23'te yavaşla; acele edilecek tek yer orası

değil.

- Dokunulmayacak slaytlar: **3, 5, 12, 14, 16, 18, 23.** İskelet bunlar.
- Köşeli parantezli yönergelerde gerçekten ekrana dön ve işaret et.